

题目

化合物 XYZ_3 是一种生活中常见的物质，其在自然界有矿石分布，其晶体结构在堆积形式上类似于 NaCl 晶体，但其正当晶胞的选取方式与 NaCl 晶体不同。若其按与 NaCl 相同的堆积方式进行堆积，则此情况下的正当晶胞体积是其实际正当晶胞的 $2/3$ 。 XYZ_3 中 X 由 6 个 Z 等距离配位； YZ_3^{n-} 平行于 ab 平面。

若 XYZ_3 的正当晶胞以 X^{n+} 为顶点，写出该晶胞中所有 Y 的原子坐标。

A. $(\frac{2}{3}, \frac{1}{3}, \frac{1}{12}), (0, 0, \frac{1}{4}), (\frac{1}{3}, \frac{2}{3}, \frac{5}{12}), (\frac{2}{3}, \frac{1}{3}, \frac{7}{12}), (0, 0, \frac{3}{4}), (\frac{1}{3}, \frac{2}{3}, \frac{11}{12})$

或

$(\frac{2}{3}, \frac{1}{3}, \frac{11}{12}), (0, 0, \frac{3}{4}), (\frac{1}{3}, \frac{2}{3}, \frac{7}{12}), (\frac{2}{3}, \frac{1}{3}, \frac{5}{12}), (0, 0, \frac{1}{4}), (\frac{1}{3}, \frac{2}{3}, \frac{1}{12})$

B. $(0, 0, 0), (\frac{1}{3}, \frac{2}{3}, \frac{1}{6}), (\frac{2}{3}, \frac{1}{3}, \frac{1}{3}), (0, 0, \frac{1}{2}), (\frac{1}{3}, \frac{2}{3}, \frac{2}{3}), (\frac{2}{3}, \frac{1}{3}, \frac{5}{6})$

C. $(0, 0, 0), (\frac{1}{3}, \frac{2}{3}, \frac{5}{6}), (\frac{2}{3}, \frac{1}{3}, \frac{2}{3}), (0, 0, \frac{1}{2}), (\frac{1}{3}, \frac{2}{3}, \frac{1}{3}), (\frac{2}{3}, \frac{1}{3}, \frac{1}{6})$

D. $(\frac{3}{4}, \frac{3}{4}, 0), (\frac{3}{4}, 0, \frac{3}{4}), (0, \frac{3}{4}, \frac{3}{4}), (\frac{3}{4}, \frac{3}{4}, \frac{3}{4})$

E. $(0, 0, 0), (\frac{1}{3}, \frac{2}{3}, \frac{1}{6}), (\frac{2}{3}, \frac{1}{3}, \frac{1}{3}), (0, 0, \frac{1}{2}), (\frac{1}{3}, \frac{2}{3}, \frac{2}{3}), (\frac{2}{3}, \frac{1}{3}, \frac{5}{6})$

F. $(\frac{2}{3}, \frac{1}{3}, \frac{1}{12}), (0, 0, \frac{1}{4}), (\frac{1}{3}, \frac{2}{3}, \frac{5}{12}), (\frac{2}{3}, \frac{1}{3}, \frac{7}{12}), (0, 0, \frac{3}{4}), (\frac{1}{3}, \frac{2}{3}, \frac{11}{12})$

G. $(\frac{2}{3}, \frac{1}{3}, \frac{11}{12}), (0, 0, \frac{3}{4}), (\frac{1}{3}, \frac{2}{3}, \frac{7}{12}), (\frac{2}{3}, \frac{1}{3}, \frac{5}{12}), (0, 0, \frac{1}{4}), (\frac{1}{3}, \frac{2}{3}, \frac{1}{12})$

答案

正确答案: A

详细解析

NaCl结构的典型特征为两种离子分别按照ccp形式堆积，因此在 XYZ_3 中也应存在同样的堆积形式。

CHECKPOINT

1 PTS

XYZ_3 存在ccp堆积

NaCl的正当晶胞属于立方晶系，有4个Na和4个Cl，而其堆积方式的一个循环，也即AcBaA型堆积（含有2个Na和2个Cl）的晶胞体积是NaCl晶胞体积的0.5倍。因此对于 XYZ_3 的正当晶胞，若体积要满足按NaCl堆积形式的正当晶胞体积的1.5倍（若其按与NaCl相同的堆积方式进行堆积，则此情况下的正当晶胞体积是其实正当晶胞的2/3），则需要有3个AcBaA大小的晶胞重合得到。结合对称性要求，应为AbC'aBcA'bCaB'c。

CHECKPOINT

1 PTS

AbC'aBcA'bCaB'c

Y原子本身没有取向，且填入的是X原子的八面体空隙

CHECKPOINT

1 PTS

Y 原子填入 X 原子的八面体空隙

并且根据B、C层位置选择的不同，可以有两套坐标。

CHECKPOINT

1 PTS

两套坐标

因此，坐标为 $(\frac{2}{3}, \frac{1}{3}, \frac{1}{12})$, $(0, 0, \frac{1}{4})$, $(\frac{1}{3}, \frac{2}{3}, \frac{5}{12})$, $(\frac{2}{3}, \frac{1}{3}, \frac{7}{12})$, $(0, 0, \frac{3}{4})$, $(\frac{1}{3}, \frac{2}{3}, \frac{11}{12})$

或

$(\frac{2}{3}, \frac{1}{3}, \frac{11}{12})$, $(0, 0, \frac{3}{4})$, $(\frac{1}{3}, \frac{2}{3}, \frac{7}{12})$, $(\frac{2}{3}, \frac{1}{3}, \frac{5}{12})$, $(0, 0, \frac{1}{4})$, $(\frac{1}{3}, \frac{2}{3}, \frac{1}{12})$

CHECKPOINT

2 PTS

坐标为 $(\frac{2}{3}, \frac{1}{3}, \frac{1}{12})$, $(0, 0, \frac{1}{4})$, $(\frac{1}{3}, \frac{2}{3}, \frac{5}{12})$, $(\frac{2}{3}, \frac{1}{3}, \frac{7}{12})$, $(0, 0, \frac{3}{4})$, $(\frac{1}{3}, \frac{2}{3}, \frac{11}{12})$ 或 $(\frac{2}{3}, \frac{1}{3}, \frac{11}{12})$, $(0, 0, \frac{3}{4})$, $(\frac{1}{3}, \frac{2}{3}, \frac{7}{12})$, $(\frac{2}{3}, \frac{1}{3}, \frac{5}{12})$, $(0, 0, \frac{1}{4})$, $(\frac{1}{3}, \frac{2}{3}, \frac{1}{12})$